# Value-Based RL

# CS 185/285

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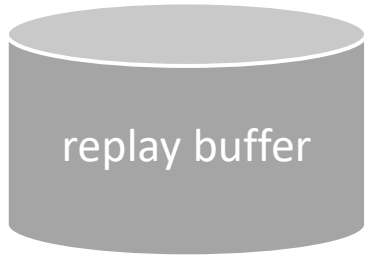
# Part 1:
# Actor-critic without the actor

# Review: actor-critic with a replay buffer

off-policy actor-critic algorithm:

1. get $(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i)$ by taking one step with $\mathbf{a} \sim \pi_\theta(\mathbf{a}|\mathbf{s})$
2. evaluate $y_i = r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \mathbb{E}_{\mathbf{a}' \sim \pi_\theta(\mathbf{a}'|\mathbf{s}'_i)}[\hat{Q}^\pi_\phi(\mathbf{s}'_i, \mathbf{a}')]$
3. update $\phi$ using $\nabla_\phi \sum_{i=1}^{B} \|\hat{Q}^\pi_\phi(\mathbf{s}_i, \mathbf{a}_i) - y_i\|^2$
4. ~~$\nabla_\theta J(\theta) \approx \sum_i \mathbb{E}_{\mathbf{a} \sim \pi_\theta(\mathbf{a}^\pi|\mathbf{s}_i)}[\nabla_\theta \log \pi_\theta(\mathbf{a}|\mathbf{s}_i) \hat{Q}^\pi_\phi(\mathbf{s}_i, \mathbf{a})]$~~
5. ~~$\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$~~

store $\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i$




use batch $\{(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i)\}_{i=1}^{B}$ from buffer

$$y_i = r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \mathbb{E}_{\mathbf{a}' \sim \pi_\theta(\mathbf{a}'|\mathbf{s}'_i)}[\hat{Q}^\pi_\phi(\mathbf{s}'_i, \mathbf{a}')]$$

$$r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \max_{\mathbf{a}'} \hat{Q}^\pi_\phi(\mathbf{s}'_i, \mathbf{a}')$$

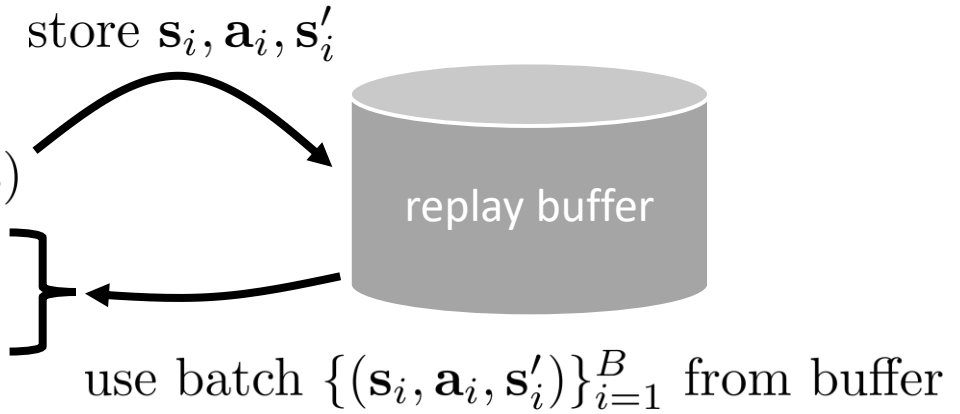
a funny idea:

$$\text{set} \quad \pi_\theta(\mathbf{a}|\mathbf{s}) = \begin{cases} 1 \text{ if } \mathbf{a} = \arg\max \hat{Q}^\pi_\phi(\mathbf{s}, \mathbf{a}) \\ 0 \text{ otherwise} \end{cases}$$

# The resulting algorithm

off-policy "critic" algorithm (Q-learning):

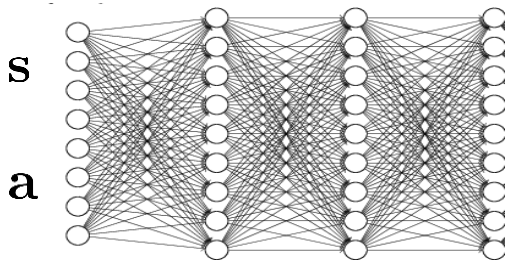
1. get $(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i)$ by taking one step with $\mathbf{a} \sim \pi_\theta(\mathbf{a}|\mathbf{s})$
2. evaluate $y_i = r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \max_{\mathbf{a}'} Q_\theta(\mathbf{s}'_i, \mathbf{a}')]$
3. update $\theta$ using $\nabla_\theta \sum_{i=1}^{B} \|Q_\theta(\mathbf{s}_i, \mathbf{a}_i) - y_i\|^2$




$$\pi_\theta(\mathbf{a}|\mathbf{s}) = \begin{cases} 1 \text{ if } \mathbf{a} = \arg\max Q_\theta(\mathbf{s}, \mathbf{a}) \\ 0 \text{ otherwise} \end{cases}$$

We could just stop here

But it helps to understand more…




$Q_\theta(\mathbf{s}, \mathbf{a})$

parameters $\theta$

# Part 2:
# Policy iteration and dynamic programming

解决确定性环境的最优决策问题
提供小规模离散场景的高效求解方案

# Policy iteration

策略迭代是通过「评估当前策略→优化策略」的循环，逐步逼近最优策略

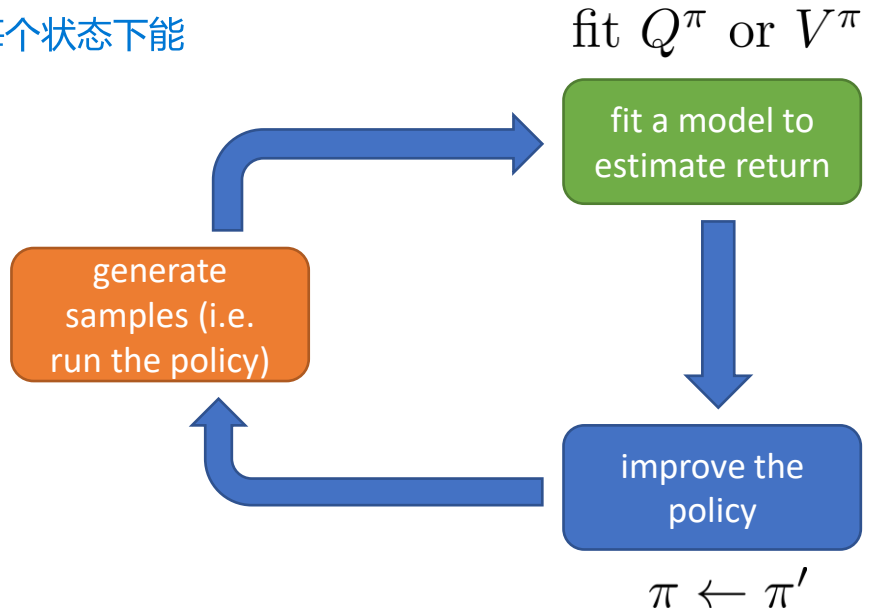
原则：基于评估得到的Q值，选择每个状态下能最大化Q值的动作，生成新策略π′

## High level idea:

policy iteration algorithm:

1. evaluate $Q^\pi(\mathbf{s}, \mathbf{a})$ ⟵ how to do this?
2. set $\pi \leftarrow \pi'$

策略更新：将当前策略替换为新策略，即π←π′，然后回到策略评估步骤，继续迭代




$$\pi'(\mathbf{a}_t|\mathbf{s}_t) = \begin{cases} 1 \text{ if } \mathbf{a}_t = \arg\max_{\mathbf{a}_t} Q^\pi(\mathbf{s}_t, \mathbf{a}_t) \\ 0 \text{ otherwise} \end{cases}$$

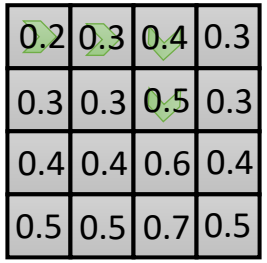
as before: $Q^\pi(\mathbf{s}, \mathbf{a}) = r(\mathbf{s}, \mathbf{a}) + \gamma E[V^\pi(\mathbf{s}')]$

let's evaluate $V^\pi(\mathbf{s})$!

不再用梯度来更新策略，而是用最大化价值的action决定的下一步策略来更新策略

# Dynamic programming

Let's assume we know $p(\mathbf{s}'|\mathbf{s},\mathbf{a})$, and $\mathbf{s}$ and $\mathbf{a}$ are both discrete (and small)




16 states, 4 actions per state

can store full $V^\pi(\mathbf{s})$ in a table!

$\mathcal{T}$ is $16 \times 16 \times 4$ tensor transition tensor

bootstrapped update: $V^\pi(\mathbf{s}) \leftarrow E_{\mathbf{a}\sim\pi(\mathbf{a}|\mathbf{s})}[r(\mathbf{s},\mathbf{a}) + \gamma E_{\mathbf{s}'\sim p(\mathbf{s}'|\mathbf{s},\mathbf{a})}[V^\pi(\mathbf{s}')]]$

just use the current estimate here

$$\pi'(\mathbf{a}_t|\mathbf{s}_t) = \begin{cases} 1 \text{ if } \mathbf{a}_t = \arg\max_{\mathbf{a}_t} Q^\pi(\mathbf{s}_t,\mathbf{a}_t) \\ 0 \text{ otherwise} \end{cases} \longrightarrow \text{deterministic policy } \pi(\mathbf{s}) = \mathbf{a}$$

更新策略方式：在状态st下，只保留能最大化动作价值的动作，其他动作的选择概率设为0，生成新的确定性策略

simplified: $V^\pi(\mathbf{s}) \leftarrow r(\mathbf{s},\pi(\mathbf{s})) + \gamma E_{\mathbf{s}'\sim p(\mathbf{s}'|\mathbf{s},\pi(\mathbf{s}))}[V^\pi(\mathbf{s}')]$

# Policy iteration with dynamic programming

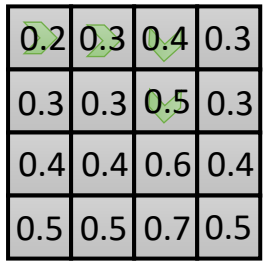
policy iteration:

1. evaluate $V^\pi(\mathbf{s})$
2. set $\pi \leftarrow \pi'$

$$\pi'(\mathbf{a}_t|\mathbf{s}_t) = \begin{cases} 1 \text{ if } \mathbf{a}_t = \arg\max_{\mathbf{a}_t} Q^\pi(\mathbf{s}_t, \mathbf{a}_t) \\ 0 \text{ otherwise} \end{cases}$$

policy evaluation:

$$V^\pi(\mathbf{s}) \leftarrow r(\mathbf{s}, \pi(\mathbf{s})) + \gamma E_{\mathbf{s}' \sim p(\mathbf{s}'|\mathbf{s}, \pi(\mathbf{s}))}[V^\pi(\mathbf{s}')]$$




16 states, 4 actions per state

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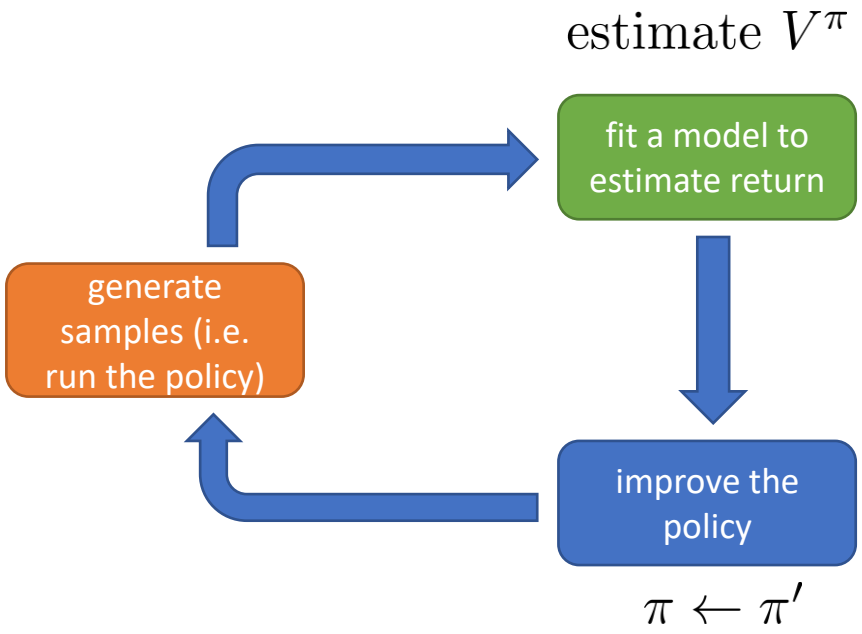



该方法依赖动态规划的特性，仅适用于以下场景：
状态s和动作a均为离散且规模较小
已知精确的状态转移概率

# Even simpler dynamic programming

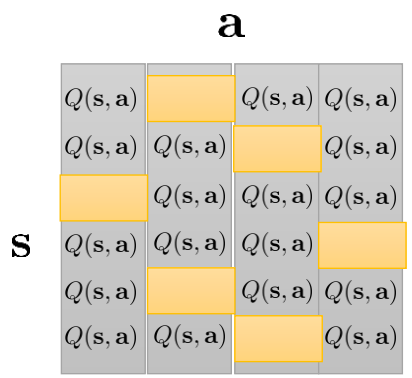
$$\pi'(\mathbf{a}_t|\mathbf{s}_t) = \begin{cases} 1 \text{ if } \mathbf{a}_t = \arg\max_{\mathbf{a}_t} Q^\pi(\mathbf{s}_t, \mathbf{a}_t) \\ 0 \text{ otherwise} \end{cases}$$

let's evaluate $Q^\pi(\mathbf{s})$!

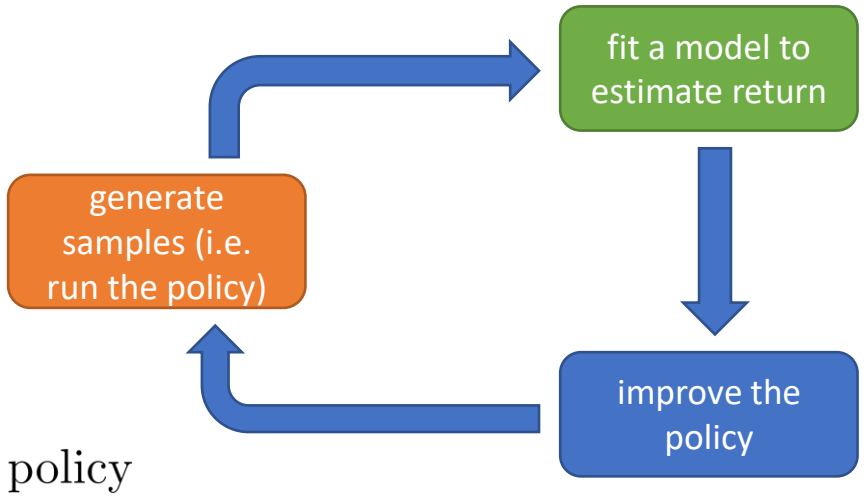
skip the policy and compute values directly!

value iteration algorithm:

1. set $Q(\mathbf{s}, \mathbf{a}) \leftarrow r(\mathbf{s}, \mathbf{a}) + \gamma E[V(\mathbf{s}')]$
2. set $V(\mathbf{s}) \leftarrow \max_{\mathbf{a}} Q(\mathbf{s}, \mathbf{a})$




$$Q^\pi(\mathbf{s}, \mathbf{a}) \leftarrow r(\mathbf{s}, \mathbf{a}) + \gamma E_{\mathbf{s}' \sim p(\mathbf{s}'|\mathbf{s},\mathbf{a})}[V^\pi(\mathbf{s}')]$$




$$\arg\max_{\mathbf{a}} Q(\mathbf{s}, \mathbf{a}) \rightarrow \text{policy}$$

$$V^\pi(\mathbf{s}) \leftarrow \max_{\mathbf{a}} Q^\pi(\mathbf{s}, \mathbf{a})$$

approximates the new value!

传统策略迭代需要「评估策略→优化策略」的循环，而该方法直接聚焦价值函数计算，通过迭代更新动作价值和状态价值，最终从Qπ中直接提取最优策略，无需显式维护策略π。

# Part 3:
# Fitted value iteration

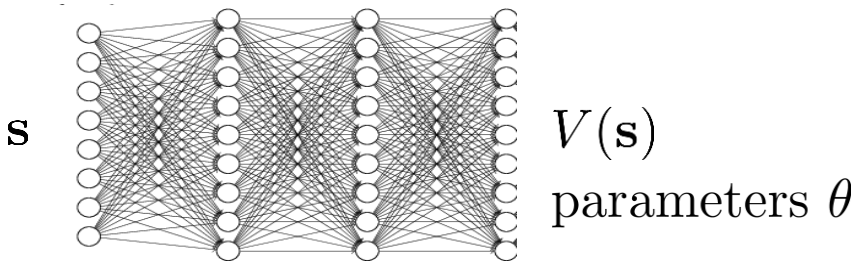
# Fitted value iteration

针对的是高维状态空间的强化学习问题

how do we represent $V(\mathbf{s})$?

big table, one entry for each discrete $\mathbf{s}$

neural net function $V : \mathcal{S} \rightarrow \mathbb{R}$




$$\mathcal{L}(\theta) = \frac{1}{2} \left\| V_\theta(\mathbf{s}) - \max_{\mathbf{a}} Q^\pi(\mathbf{s}, \mathbf{a}) \right\|^2$$

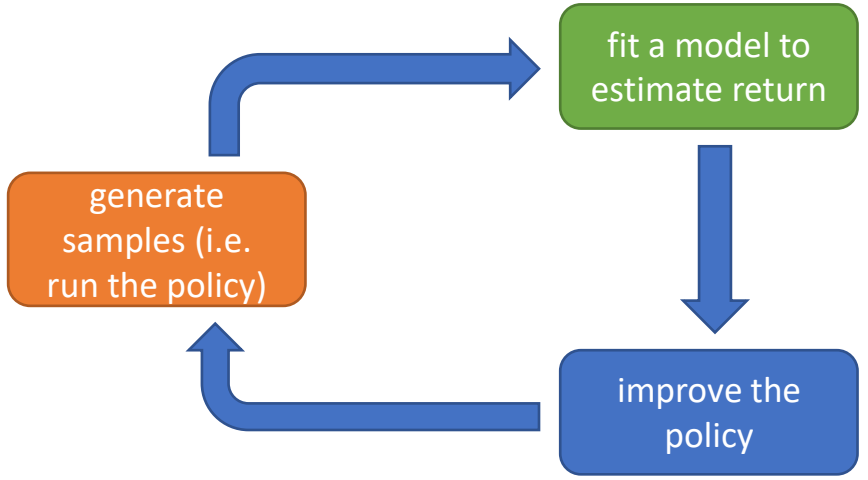
$$\mathbf{s} = 0 : V(\mathbf{s}) = 0.2$$
$$\mathbf{s} = 1 : V(\mathbf{s}) = 0.3$$
$$\mathbf{s} = 2 : V(\mathbf{s}) = 0.5$$


$$|\mathcal{S}| = (255^3)^{200 \times 200}$$

(more than atoms in the universe)

curse of dimensionality

$$Q^\pi(\mathbf{s}, \mathbf{a}) \leftarrow r(\mathbf{s}, \mathbf{a}) + \gamma E_{\mathbf{s}' \sim p(\mathbf{s}'|\mathbf{s},\mathbf{a})}[V^\pi(\mathbf{s}')]$$




$$V^\pi(\mathbf{s}) \leftarrow \max_{\mathbf{a}} Q^\pi(\mathbf{s}, \mathbf{a})$$

针对一个极高维的连续状态空间，采用动态规划方案和蒙特卡洛方法迭代的表格存储值函数是不现实的。所以用神经网络对s->值函数V的映射进行拟合

fitted value iteration algorithm:

1. set $y_i \leftarrow \max_{\mathbf{a}_i} (r(\mathbf{s}_i, \mathbf{a}_i) + \gamma E[V_\theta(\mathbf{s}'_i)])$
2. set $\theta \leftarrow \arg\min_\theta \frac{1}{2} \sum_i \left\| V_\theta(\mathbf{s}_i) - y_i \right\|^2$

# What if we don't know the transition dynamics?

fitted value iteration algorithm:

1. set $y_i \leftarrow \max_{\mathbf{a}_i}(r(\mathbf{s}_i, \mathbf{a}_i) + \gamma E[V_\theta(\mathbf{s}'_i)])$
2. set $\theta \leftarrow \arg\min_\theta \frac{1}{2}\sum_i \|V_\theta(\mathbf{s}_i) - y_i\|^2$

该算法的核心是通过样本拟合值函数，无需显式建模状态转移概率
该方法仍需知道不同动作的执行结果，可通过历史样本或交互数据近似期望项。

need to know outcomes for different actions!

依赖于：1. 要么已知状态转移概率分布 2. 要么通过sampling拟合，但是有偏

Back to policy iteration…

policy iteration:

1. evaluate $Q^\pi(\mathbf{s}, \mathbf{a})$
2. set $\pi \leftarrow \pi'$

$$\pi'(\mathbf{a}_t|\mathbf{s}_t) = \begin{cases} 1 \text{ if } \mathbf{a}_t = \arg\max_{\mathbf{a}_t} Q^\pi(\mathbf{s}_t, \mathbf{a}_t) \\ 0 \text{ otherwise} \end{cases}$$

policy evaluation:

$$V^\pi(\mathbf{s}) \leftarrow r(\mathbf{s}, \pi(\mathbf{s})) + \gamma E_{\mathbf{s}' \sim p(\mathbf{s}'|\mathbf{s}, \pi(\mathbf{s}))}[V^\pi(\mathbf{s}')]$$

$$Q^\pi(\mathbf{s}, \mathbf{a}) \leftarrow r(\mathbf{s}, \mathbf{a}) + \gamma E_{\mathbf{s}' \sim p(\mathbf{s}'|\mathbf{s}, \mathbf{a})}[Q^\pi(\mathbf{s}', \pi(\mathbf{s}'))]$$

can fit this using samples

# Can we do the “max” trick again?

拟合Q迭代（Fitted Q Iteration）算法：

相比策略迭代：策略迭代需要显式维护策略，且依赖完整的状态转移模型来完成策略评估和更新；拟合Q迭代无需显式策略，也不需要已知状态转移概率，仅通过样本数据就能完成学习。

相比拟合值迭代：拟合值迭代需要通过“max”操作隐式优化策略，且在计算未来回报期望时，通常需要依赖状态转移模型或大量采样；拟合Q迭代通过用下一个状态的最大Q值近似未来回报期望，完全不需要模拟动作执行，也不依赖状态转移模型。

policy iteration:

1. evaluate $V^\pi(\mathbf{s})$
2. set $\pi \leftarrow \pi'$

fitted value iteration algorithm:

1. set $y_i \leftarrow \max_{\mathbf{a}_i}(r(\mathbf{s}_i, \mathbf{a}_i) + \gamma E[V_\theta(\mathbf{s}'_i)])$
2. set $\theta \leftarrow \arg\min_\theta \frac{1}{2}\sum_i \|V_\theta(\mathbf{s}_i) - y_i\|^2$

no policy, compute value directly

can we do this with Q-values **also**, without knowing the transitions?

这一点和actor-critc算法很不一样

doesn’t require simulation of actions!

fitted Q iteration algorithm:

1. set $y_i \leftarrow r(\mathbf{s}_i, \mathbf{a}_i) + \gamma E[V_\theta(\mathbf{s}'_i)]$ ← approxiate $E[V(\mathbf{s}'_i)] \approx \max_{\mathbf{a}'} Q_\theta(\mathbf{s}'_i, \mathbf{a}'_i)$
2. set $\theta \leftarrow \arg\min_\theta \frac{1}{2}\sum_i \|Q_\theta(\mathbf{s}_i, \mathbf{a}_i) - y_i\|^2$

+ works even for off-policy samples 支持离策略样本（off-policy samples）：可以利用非当前策略产生的历史数据进行学习，数据利用率更高

+ only one network, no high-variance policy gradient 结构简单：仅需一个网络，避免了策略梯度方法的高方差问题

- no convergence guarantees for non-linear function approximation (more on this later)

非线性函数近似的收敛性无法保证：当用神经网络等非线性函数近似Q值时，算法可能无法收敛到最优解（后续会进一步讲解相关解决方案）

# Fitted Q-iteration

full fitted Q-iteration algorithm:

parameters

1. collect dataset $\{(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i)\}$ using some policy

dataset size $N$, collection policy

$K\times$

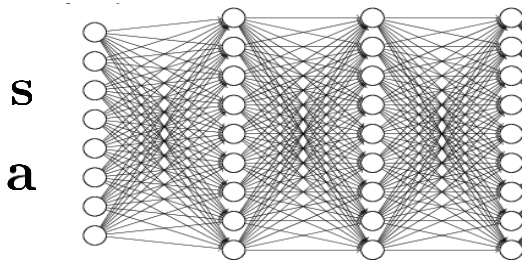
2. set $y_i \leftarrow r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \max_{\mathbf{a}'_i} Q_\theta(\mathbf{s}'_i, \mathbf{a}'_i)$

iterations $K$

3. set $\theta \leftarrow \arg\min_\theta \frac{1}{2} \sum_i \|Q_\theta(\mathbf{s}_i, \mathbf{a}_i) - y_i\|^2$

gradient steps $S$

train the new q function, which phi is the weight




$Q_\theta(\mathbf{s}, \mathbf{a})$

parameters $\theta$

当Q函数收敛到最优Q函数后，策略可以直接通过贪心策略从Q函数导出：π(s)为使得Q function值最大的action， 作为最终的策略函数。整个策略函数的更新都是隐式更新

# Part 4:
# Back to Q-learning

# Why is this algorithm off-policy?

不需要实时采用最新的策略函数来获得数据

full fitted Q-iteration algorithm:

1. collect dataset $\{(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i)\}$ using some policy

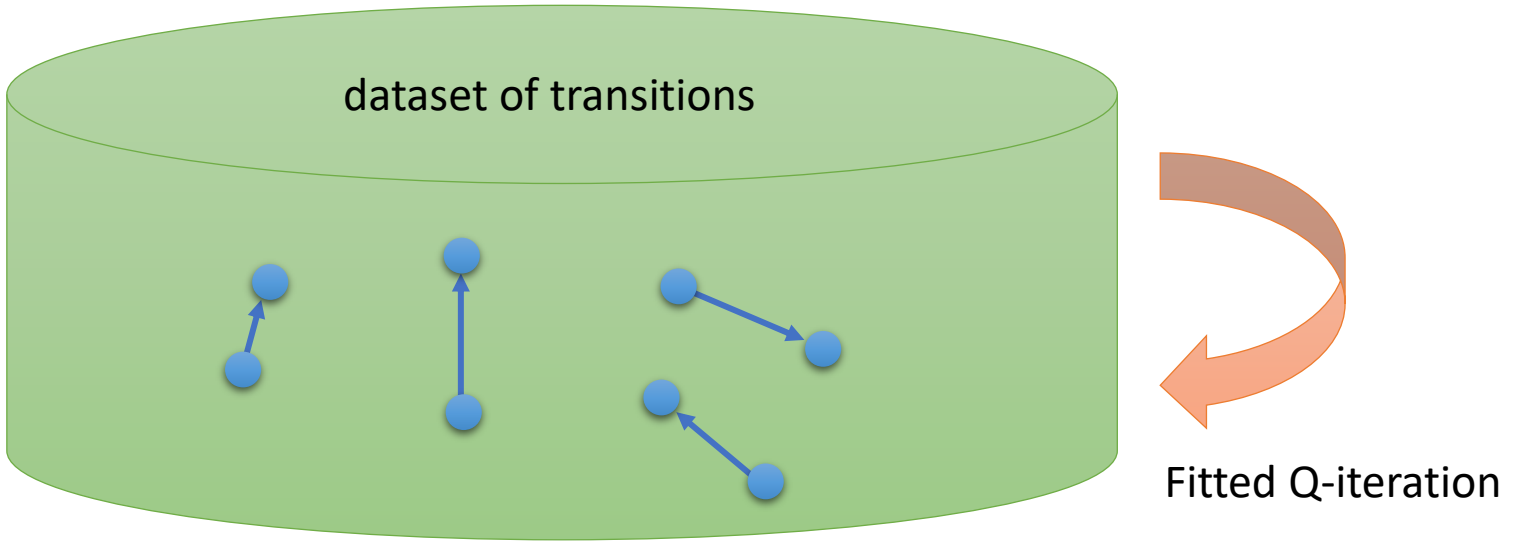
$K\times$

2. set $y_i \leftarrow r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \max_{\mathbf{a}'_i} Q_\theta(\mathbf{s}'_i, \mathbf{a}'_i)$
3. set $\theta \leftarrow \arg\min_\theta \frac{1}{2} \sum_i \|Q_\theta(\mathbf{s}_i, \mathbf{a}_i) - y_i\|^2$

given $\mathbf{s}$ *and* $\mathbf{a}$, transition is independent of $\pi$

this approximates the value of $\pi'$ at $\mathbf{s}'_i$

$$\pi'(\mathbf{a}_t|\mathbf{s}_t) = \begin{cases} 1 \text{ if } \mathbf{a}_t = \arg\max_{\mathbf{a}_t} Q^\pi(\mathbf{s}_t, \mathbf{a}_t) \\ 0 \text{ otherwise} \end{cases}$$



# What is fitted Q-iteration optimizing?

full fitted Q-iteration algorithm:

1. collect dataset $\{(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i)\}$ using some policy

$K\times$

2. set $y_i \leftarrow r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \max_{\mathbf{a}'_i} Q_\theta(\mathbf{s}'_i, \mathbf{a}'_i)$ ← this max improves the policy (tabular case)
3. set $\theta \leftarrow \arg\min_\theta \frac{1}{2} \sum_i \|Q_\theta(\mathbf{s}_i, \mathbf{a}_i) - y_i\|^2$

↑ error $\mathcal{E}$

$$\mathcal{E} = \frac{1}{2} E_{(\mathbf{s},\mathbf{a})\sim\beta} \left[ \left( Q_\theta(\mathbf{s}, \mathbf{a}) - [r(\mathbf{s}, \mathbf{a}) + \gamma \max_{\mathbf{a}'} Q_\theta(\mathbf{s}', \mathbf{a}')] \right)^2 \right]$$ 贝尔曼误差

if $\mathcal{E} = 0$, then $Q_\theta(\mathbf{s}, \mathbf{a}) = r(\mathbf{s}, \mathbf{a}) + \gamma \max_{\mathbf{a}'} Q_\theta(\mathbf{s}', \mathbf{a}')$

this is an *optimal* Q-function, corresponding to optimal policy $\pi'$:

$$\pi'(\mathbf{a}_t|\mathbf{s}_t) = \begin{cases} 1 \text{ if } \mathbf{a}_t = \arg\max_{\mathbf{a}_t} Q_\phi(\mathbf{s}_t, \mathbf{a}_t) \\ 0 \text{ otherwise} \end{cases}$$

maximizes reward

sometimes written $Q^\star$ and $\pi^\star$

most guarantees are lost when we leave the tabular case (e.g., use neural networks)

当我们从传统的表格数据（tabular data）建模转向使用神经网络等复杂模型时，许多统计方法中存在的理论保障就不再成立了。例如：可解释性、参数显著性、收敛性证明、误差分布假设等

# Online Q-learning algorithms

这两种算法都支持离策略学习，意味着学习过程使用的数据可以和当前策略无关，能更灵活地探索环境。
两种算法都遵循"生成样本（执行策略）→ 拟合模型（估计回报）→ 优化策略"的循环。

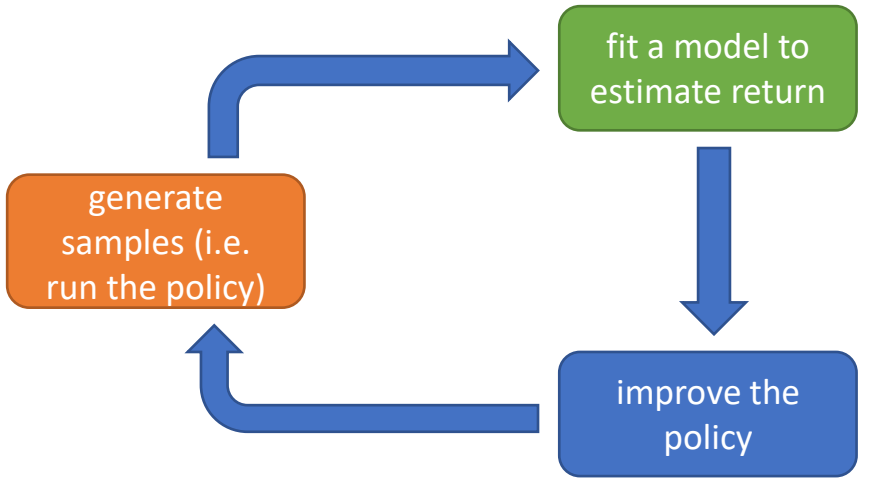
$$Q_\theta(\mathbf{s},\mathbf{a}) \leftarrow r(\mathbf{s},\mathbf{a}) + \gamma \max_{\mathbf{a}'} Q_\theta(\mathbf{s}',\mathbf{a}')$$

完全基于批量数据的Q学习方法

full fitted Q-iteration algorithm:

1. collect dataset $\{(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i)\}$ using some policy
2. set $y_i \leftarrow r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \max_{\mathbf{a}'_i} Q_\theta(\mathbf{s}'_i, \mathbf{a}'_i)$
3. set $\theta \leftarrow \arg\min_\theta \frac{1}{2} \sum_i \|Q_\theta(\mathbf{s}_i, \mathbf{a}_i) - y_i\|^2$

$K\times$




$$\mathbf{a} = \arg\max_{\mathbf{a}} Q_\theta(\mathbf{s},\mathbf{a})$$

基于单样本在线更新的Q学习方法

off policy, so many choices here!

online Q iteration algorithm:

1. take some action $\mathbf{a}_i$ and observe $(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i)$
2. $y_i = r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \max_{\mathbf{a}'} Q_\theta(\mathbf{s}'_i, \mathbf{a}'_i)$
3. $\theta \leftarrow \theta - \alpha \frac{dQ_\theta}{d\theta}(\mathbf{s}_i, \mathbf{a}_i)(Q_\theta(\mathbf{s}_i, \mathbf{a}_i) - y_i)$

对Q function 求导，通过梯度下降法更新Q function reward预测模型参数theta

只有当θ训练得足够准确，Q函数能精准预测长期回报时，才能通过这个公式得到真正能最大化回报的最优动作

# Exploration with Q-learning

online Q iteration algorithm:

1. take some action $\mathbf{a}_i$ and observe $(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i)$
2. $y_i = r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \max_{\mathbf{a}'} Q_\theta(\mathbf{s}'_i, \mathbf{a}'_i)$
3. $\theta \leftarrow \theta - \alpha \frac{dQ_\theta}{d\theta}(\mathbf{s}_i, \mathbf{a}_i)(Q_\theta(\mathbf{s}_i, \mathbf{a}_i) - y_i)$

final policy:

$$\pi(\mathbf{a}_t|\mathbf{s}_t) = \begin{cases} 1 \text{ if } \mathbf{a}_t = \arg\max_{\mathbf{a}_t} Q_\theta(\mathbf{s}_t, \mathbf{a}_t) \\ 0 \text{ otherwise} \end{cases}$$

why is this a bad idea for step 1?

如果在第一步就直接选择最大化奖励的action，会导致探索不足。算法会一直重复执行当前认为最优的动作，完全不会尝试其他可能带来更高长期回报的动作，容易陷入局部最优解。

$$\pi(\mathbf{a}_t|\mathbf{s}_t) = \begin{cases} 1 - \epsilon \text{ if } \mathbf{a}_t = \arg\max_{\mathbf{a}_t} Q_\theta(\mathbf{s}_t, \mathbf{a}_t) \\ \epsilon/(|\mathcal{A}| - 1) \text{ otherwise} \end{cases}$$

“epsilon-greedy”

以1−ϵ的概率选择当前Q值最高的动作（利用已学知识），以ϵ的概率随机选择其他动作（探索未知）。ϵ的大小可以灵活调整，比如训练初期设大值多探索，后期设小值多利用。

$$\pi(\mathbf{a}_t|\mathbf{s}_t) \propto \exp(Q_\theta(\mathbf{s}_t, \mathbf{a}_t))$$

“Boltzmann exploration”

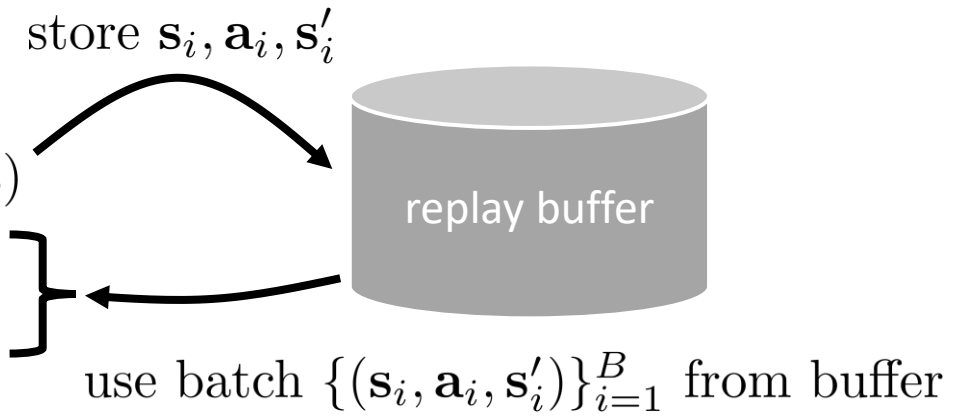
动作的选择概率和其Q值的指数成正比，Q值越高的动作被选中的概率越大，但所有动作都有被选中的可能。探索的随机性更平滑，不会像ϵ-贪心那样在探索和利用之间做硬切换，适合动作空间连续的场景。

We’ll discuss exploration in detail in a later lecture!

# Bringing it back together

Q-learning with replay buffer:

1. get $(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i)$ by taking one step with $\mathbf{a} \sim \pi_\theta(\mathbf{a}|\mathbf{s})$
2. evaluate $y_i = r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \max_{\mathbf{a}'} Q_\theta(\mathbf{s}'_i, \mathbf{a}')]$
3. update $\theta$ using $\nabla_\theta \sum_{i=1}^{B} \|Q_\theta(\mathbf{s}_i, \mathbf{a}_i) - y_i\|^2$




$$\pi_\theta(\mathbf{a}|\mathbf{s}) = \begin{cases} 1-\epsilon \text{ if } \mathbf{a} = \arg\max_{\mathbf{a}} Q_\theta(\mathbf{s}, \mathbf{a}) \\ \epsilon/(|\mathcal{A}|-1) \text{ otherwise} \end{cases}$$

If you try to code this up, it almost certainly won't work

We'll find out how to make it work next time!